
1 Concept of the Code

The main idea of the script is **dataset validation**:

1. **Load a dataset** (CSV) into Python.
2. **Iterate through each row** and send the text to **TrustGuard** for validation.
3. **TrustGuard checks** whether the text is safe/valid according to a schema.
4. **Mark each row** as "Approved" or "Rejected" depending on TrustGuard's result.
5. **Inspect rejected rows** to see why they failed.
6. **Get summary statistics** using TrustGuard's built-in stats.
7. **Save the updated dataset** with the approval status.

So conceptually:

Raw Text CSV → JSON → TrustGuard → Approved/Rejected → Save CSV

2 What is Pandas?

Pandas is a **Python library for data manipulation and analysis**. It allows you to:

- Load datasets from CSV, Excel, or SQL databases.
- Handle tabular data with **DataFrames** (like Excel sheets in Python).
- Filter, group, sort, or modify data easily.
- Work efficiently even with millions of rows.

In your code:

- `df = pd.read_csv("Sentiment140_clean.csv")` → loads the CSV into a Pandas DataFrame.
 - `df["sent"] = approval_list` → adds a new column with approval status.
 - `df[df["sent"] == "Rejected"]` → filters rows that were rejected.
 - `df.to_csv(...)` → saves the DataFrame back to CSV.
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3 What is TrustGuard?

TrustGuard is a **data validation service** (or library) that ensures your content meets **specific rules or schemas**.

In your script:

- `guard = TrustGuard(schema_class=GenericResponse)` → initializes TrustGuard with a generic validation schema.
- `guard.validate(json_text)` → checks if the text passes the validation rules.
- `result.is_approved` → boolean that tells you whether the text is **safe/valid**.
- `guard.get_stats()` → provides overall validation statistics like total rows approved or rejected.

Think of TrustGuard as a **content safety or quality checker** for your dataset.

4 How Pandas + TrustGuard work together

1. **Pandas** handles the data: reading CSV, storing rows, filtering, and saving results.
 2. **TrustGuard** validates each row's content.
 3. After validation, Pandas stores the results in a new column (`sent`), so you can analyze, filter, or save it.
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5 Conceptual Flow in Your Code

1. **Load Data** → `df = pd.read_csv(...)`
2. **Loop each row** → `for _, row in df.iterrows()`
3. **Prepare JSON** → `json_payload = {...}; json.dumps(json_payload)`
4. **Validate with TrustGuard** → `result = guard.validate(json_text)`
5. **Mark Approval** → `if result.is_approved: "Approved" else "Rejected"`

6. **Add Column** → `df["sent"] = approval_list`
 7. **Inspect & Analyze** → `df[df["sent"] == "Rejected"], guard.get_stats()`
 8. **Save Dataset** → `df.to_csv(...)`
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In short:

- **Pandas** = the data handler
 - **TrustGuard** = the content validator
 - **The script** = the bridge between raw data and clean, validated data
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